

MediAI: Codebase Audit, Market Analysis and Feature Roadmap

An evidence-based assessment of what the project currently does, how it sits against shipping products in the Indian and global markets, and a prioritised set of features designed to make it a defensible AI systems project.

Prepared from a full source-code audit of the MediAI repository, a review of six consumer symptom-checking products, the Indian EMR market, and five 2026 medical-AI benchmark papers.

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1. What the project claims versus what the code does

Summary of findings

The project is not small, but every AI behaviour in it is a network call to a third-party hosted model. There is no model in the repository that was trained, tuned or evaluated by the team. That is the substantive weakness, and it is fixable.

0

Models trained by the team

3 of 5

Risk models actually working

1

Messages of context sent to the LLM

0

Evaluation metrics measured

CLAIMED	WHAT ACTUALLY HAPPENS	EVIDENCE
CNN for medical image scanning	No CNN exists. Images are base64-encoded and sent to the Gemini vision API. No TensorFlow, PyTorch, Keras or ONNX model or weights anywhere in the repository.	<code>client/src/lib/aiService.ts:259-324</code>
5 ML models for disease prediction	Three work (diabetes, heart, liver). Kidney returns HTTP 501. All three are remote Hugging Face Gradio Spaces called over the network, not models the team trained or serves.	<code>ml_service/main.py:45-79, 629-633</code>
Explainable risk factors with weights	The weights are hardcoded literals inside if/else branches — glucose above 125 yields a fixed 0.35. No SHAP, no LIME, no feature importances from any model.	<code>ml_service/main.py:211-562</code>
Risk percentage from the model	The percentage is extracted with a regular expression from the Gradio Space's rendered markdown text. If that Space changes its wording, the numbers silently break.	<code>ml_service/main.py:88-182</code>
Conversational AI assistant	Each turn sends only the latest user message to Gemini. No prior turns are included, so the model is effectively stateless. This is the root cause of the assistant failing to ask coherent follow-up questions.	<code>client/src/lib/aiService.ts:138-148</code>
AI-powered analysis	Single-shot prompt and print. No retrieval, no vector store, no function calling, no tool use, no agent loop, no evaluation harness.	<code>client/src/lib/aiService.ts</code>
Encrypted, enterprise-grade security	Marketing copy only. No encryption implementation exists. The Gemini API key is compiled into the client bundle and is readable by any visitor.	<code>client/src/lib/aiService.ts:10</code>

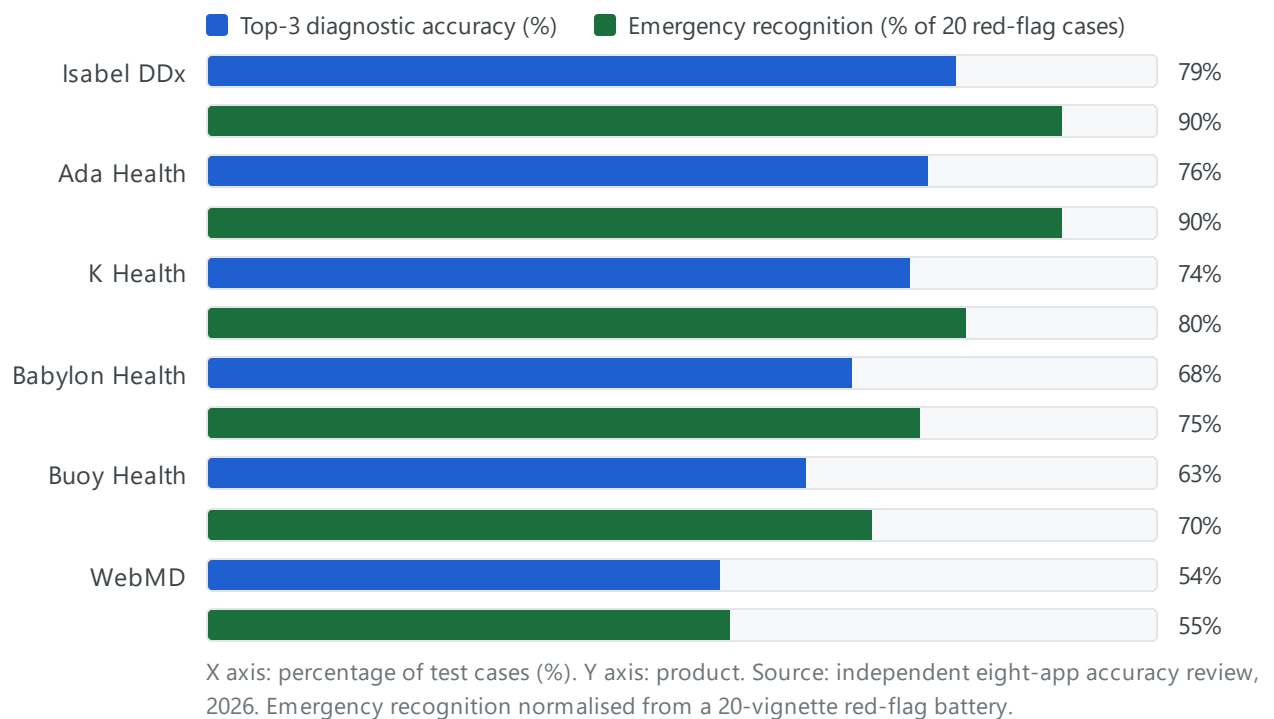
CLAIMED	WHAT ACTUALLY HAPPENS	EVIDENCE
Voice analysis	Browser Web Speech API transcription with a hardcoded <code>en-US</code> locale. A real speech SDK is present in dependencies but never imported.	<code>client/src/lib/speechService.ts:14-16</code>

Immediate security action, independent of any roadmap decision

The Gemini API key ships inside the client-side JavaScript bundle and is readable by anyone who loads the site. All model calls should be moved behind the Express server and the key rotated. The same applies to any other credentials committed to configuration files.

2. The market: consumer symptom checkers

Consumer symptom checking is a mature, crowded and regulated category. Accuracy across the leading products has plateaued in the seventies, and the products are backed by companies with clinical teams and regulatory budgets.

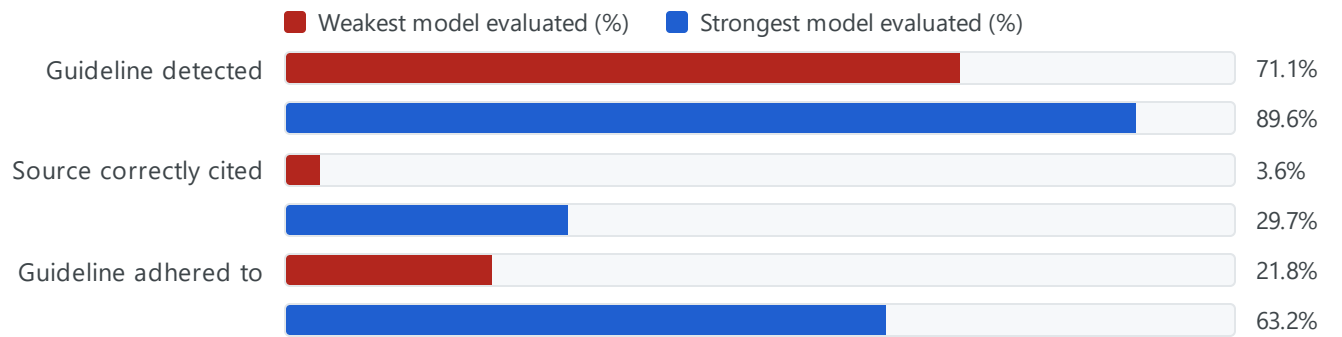


The regulatory constraint

Under EU MDR Rule 11, software that outputs a differential diagnosis or a triage decision is classified as a Class IIa medical device, and a disclaimer does not change the intended purpose. Competing on diagnostic accuracy therefore walks the project into a regulated category it cannot ship in. Competing on **verifiability** does not.

3. The research gap that is actually open

A decade-scale benchmark of clinical practice guideline adherence evaluated eight leading large language models against 32,155 clinical recommendations extracted from 3,418 guideline documents across nine countries. The result identifies a specific, unsolved failure mode.



X axis: percentage of 32,155 clinical recommendations (%). Y axis: capability stage. Source: decade-scale clinical practice guideline benchmark, 8 leading LLMs, 3,418 guideline documents, 9 countries and 2 international organisations.

Reposition the project in one sentence

Not "an AI health assistant" — those exist. Instead: **a health AI that can prove every statement it makes, ask the questions that actually reduce uncertainty, and report its own failure rate.** Anyone can call a language model. Almost nobody builds the verification layer around it, and that layer is measurable, defensible and squarely an AI systems contribution.

4. HealthPlix and the Indian EMR landscape

HealthPlix is frequently cited as a comparison point for this project. It is important to be precise about what it actually is, because the comparison is often made incorrectly.

Product category	Doctor-side electronic medical record system, not a patient-facing application
Scale	14,000+ doctors, 60M+ patients, 370+ cities
Flagship AI (H.A.L.O)	Ambient scribe: listens to the consultation and emits a structured prescription
Training data	100M+ consultations recorded inside their own EMR
Other AI	Drug-drug interaction alerts, clinical decision support, practice analytics
Languages	Prescription output in 14 regional languages
Compliance	Aligned with the Ayushman Bharat Digital Mission

What this means strategically

HealthPlix is not a competitor to a patient assistant; it is a different product class serving the doctor. The comparison is best read as an observation about where clinical value sits, not as a claim that the project already exists. Competing on ambient scribing against 100 million consultations of proprietary training data is not viable. Competing where they have no presence is.

Where the incumbents sit

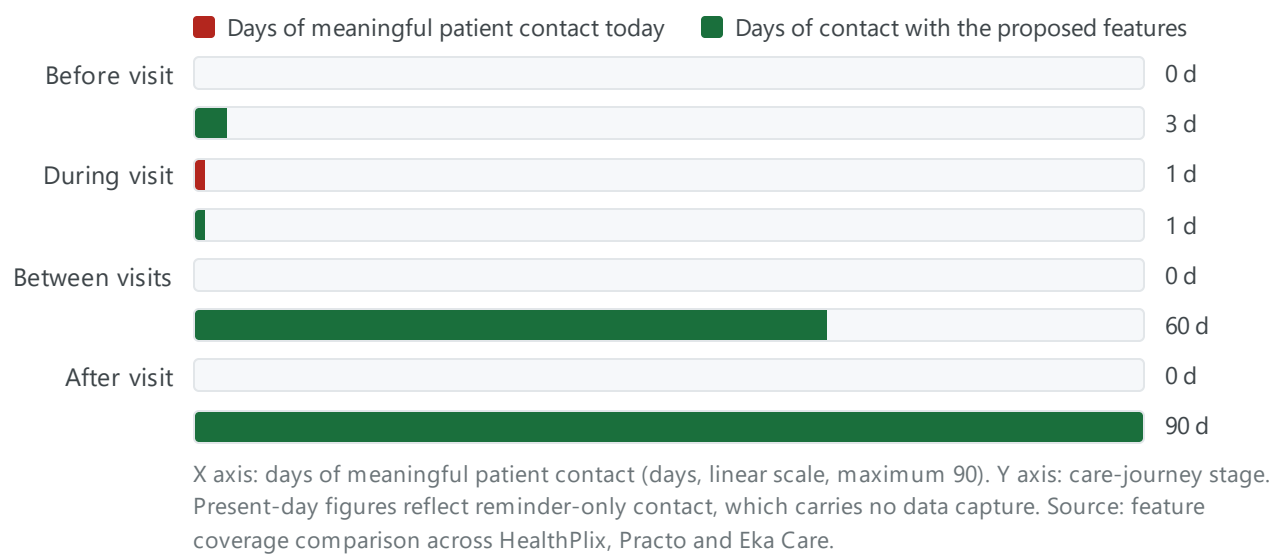
CARE-JOURNEY STAGE	HEALTHPLIX	PRACTO	EKA CARE	STATUS
Before the visit	Not covered	Discovery and booking only	Not covered	Open
During the visit	Ambient scribe, Rx, CDS	EMR and teleconsult	AI voice documentation	Crowded
Prescription written	DDI within own data	Basic	Basic	Crowded
Between visits	Reminders only	Reminders only	Reminders only	Open
Did the treatment work	Not measured	Not measured	Not measured	Open

CARE-JOURNEY STAGE	HEALTHPLIX	PRACTO	EKA CARE	STATUS
Across different doctors	Blind outside own network	Blind outside own network	Partial via ABDM	Open

The structural advantage available to a patient-side product

An EMR vendor only ever observes what happens inside its own software. An Indian patient typically consults three or four doctors across unconnected systems, takes medicines that nobody reconciles, and spends 364 days a year outside a clinic. Only the patient holds the complete record. A patient-side application can therefore build things that are structurally impossible for an EMR vendor, regardless of how much capital or data that vendor has.

5. The three uncovered windows



The entire industry competes over the single covered day. The roughly 150 uncovered days on either side of it are where the data lives, and data is the only asset that compounds.

6. Features: before the visit

Window: booking confirmed to walking in, typically hours to days. Currently the entire state of the art is a reminder notification. The patient is at home, unhurried and willing to answer questions; the doctor, who will have seven minutes, is not.

Vitals from the phone camera (remote photoplethysmography)

looks 10/10 · 3 weeks

The patient looks into the front camera for 30 seconds and the application estimates heart rate and respiratory rate from micro-variations in facial skin colour caused by blood flow.

WHY NOBODY HAS IT

It sounds as though it cannot work, so no product in the Indian consumer market ships it. The underlying signal-processing approach is well established and published, and it requires no hardware the patient does not already own.

HOW YOU BUILD IT

Detect the face, average the green channel over a forehead region across frames, detrend and bandpass the resulting one-dimensional signal to 0.7–4 Hz, then take the dominant FFT peak. The published CHROM and POS methods are a few hundred lines. Validate against an inexpensive pulse oximeter.

STACK

OpenCV or MediaPipe face mesh, SciPy signal processing, FFT

Come-prepared engine: predict the tests the doctor will order

looks 7/10 · 2 weeks

From the intake conversation, predict the laboratory tests most likely to be ordered and advise the patient to complete them beforehand, or to arrive fasting.

WHY NOBODY HAS IT

Possibly the highest practical-value item on this list for Indian patients, and entirely absent from the market. At present a patient travels to a clinic, is told to return fasting, and loses a second day and a second consultation fee.

HOW YOU BUILD IT

Retrieve the workup section of the relevant guideline for the presenting complaint and extract the recommended investigations. A retrieval and extraction task over a corpus already being built, not a prediction model.

STACK

Guideline retrieval, structured extraction, fasting-rule table

ALSO STRONG IN THIS WINDOW

WHAT IT DOES

STACK

LOOKS

WEEKS

Wrong-specialty guard

Verifies the chosen specialty matches the complaint before booking is confirmed. Self-referral to the wrong specialist is routine in India and costs a wasted fee plus weeks of delay.

Text classifier, existing booking flow

6/10

1

Symptom timeline reconstruction

Converts a rambling narrative into a chronological chart of onset, progression, aggravating factors and self-medication. Doctors think in timelines; patients speak in fragments.

Schema extraction, temporal normalisation, SVG

7/10

2

Autonomous intake agent

Interviews the patient before the appointment and produces a one-page doctor brief with a ranked differential and flagged red flags.

Agent loop, structured output, FHIR-shaped brief

7/10

2

Waiting-window escalation

Re-evaluates urgency during the wait. If a red flag appears, escalates rather than letting the patient sit on a three-day-old appointment.

Rule engine, scheduled re-evaluation

8/10

2

7. Features: between visits

Window: day two to day sixty, by far the longest. Currently only fixed-schedule medication reminders, sent identically to everyone, with no observation of whether anything worked. This is where deterioration, side effects and silent non-adherence all occur unobserved, and it is the only window long enough to produce time-series data.

Passive digital phenotyping from app interaction

looks 10/10 · 2 weeks

Infer trends in wellbeing from how the patient uses the application rather than what they report: typing speed, response latency, session times, message length and sentiment drift.

WHY NOBODY HAS IT

An active research area under the name digital phenotyping, and no Indian consumer health product implements it. The notable point is that the raw material is already being logged.

HOW YOU BUILD IT

Firestore messages already carry timestamps. Derive per-session features, establish a personal baseline, then apply trend and change-point detection. No new data collection is required.

STACK

Feature engineering over existing logs, Ruptures or CUSUM, sentiment model

Drug side-effect early warning

looks 9/10 · 2 weeks

When a new symptom is reported, check it against the known adverse-effect profile of every drug the patient recently started and flag the probable culprit.

WHY NOBODY HAS IT

Interaction checkers run at the moment of prescribing. Nobody watches for the effect appearing two weeks later, which is precisely when patients discontinue medication silently and no clinician finds out.

HOW YOU BUILD IT

SIDER is an open drug side-effect dataset. Match reported symptoms against the profiles of active drugs, weighted by time since initiation. Predominantly a data-joining problem.

STACK

SIDER dataset, RxNorm normalisation, temporal scoring

ALSO STRONG IN THIS WINDOW	WHAT IT DOES	STACK	LOOKS	WEEKS
Environmental trigger inference	Correlates the symptom log against local air quality, temperature and humidity to surface personal triggers. Air quality is a major driver of respiratory presentation in India and is free to query, yet no application personalises against it.	OpenAQ or CPCB feed, weather API, lagged cross-correlation	9/10	2
Personal baseline drift detection	Alerts on deviation from the individual's own baseline rather than a population reference range. A resting heart rate moving from 52 to 71 is alarming for that person and unremarkable on a population chart.	Bayesian online changepoint detection	8/10	2
Serial photo progression measurement	Tracks a wound, rash or swelling across repeat photographs and quantifies change in area and colour, without naming a diagnosis. Measuring change is more useful and considerably safer than classification, and avoids the regulatory problem.	OpenCV segmentation or SAM, colour-space analysis	9/10	3
Recovery trajectory versus expected curve	Plots reported severity against the typical recovery curve for the condition and flags materially slower progress. Nobody currently tells a patient whether their recovery is normal.	Curve fitting, guideline-derived duration priors	8/10	3
Proactive check-in agent	Initiates contact rather than waiting to be opened, referencing prior state. Requires genuine conversational memory, which the application currently lacks entirely.	Scheduled jobs, rolling clinical summary, push notification	7/10	2

8. Features: after the visit

Window: outcome and continuity, day seven onward. Currently nothing — no platform in this market systematically asks whether the treatment worked. Outcome data is the one asset that compounds, and no incumbent is collecting it because they lose the patient at the clinic door.

Causal treatment effectiveness across your own users

looks 10/10 · 3 weeks

For patients with similar presentations, how long did symptoms actually take to resolve under each treatment approach, adjusted so that sicker patients do not make effective treatments appear worse.

WHY NOBODY HAS IT

This is the data flywheel absent from the market, because nobody collects patient-reported outcomes after the visit. Naive averages would be misleading, and correcting for that confounding is the substantive contribution.

HOW YOU BUILD IT

Propensity score matching or inverse probability weighting over the outcome table. DoWhy and EconML implement the estimators; the work is defining treatment, outcome and confounders honestly and stating the limitations explicitly.

STACK

DoWhy or EconML, propensity matching, outcome schema

Patient-reported outcome dashboard for the doctor

looks 8/10 · 2 weeks

Shows the doctor what happened after their patients left: of the last fifty with a given presentation, what proportion reported resolution by day seven.

WHY NOBODY HAS IT

Doctors almost never receive outcome feedback. EMR analytics report volume and revenue, not recovery, because the EMR loses the patient at the door. This is also the feature that drives doctor-side adoption of the application.

HOW YOU BUILD IT

Aggregate follow-up responses per presentation. The difficult part is raising follow-up response rates, which the proactive check-in agent addresses.

STACK

Aggregation over outcome table, doctor-side view, consent model

ALSO STRONG IN THIS WINDOW	WHAT IT DOES	STACK	LOOKS	WEEKS
Second-opinion divergence detector	When a patient consults two doctors for the same problem, detects where the plans disagree on drug, dose, duration or investigation and presents the difference neutrally. Structurally impossible for an EMR vendor, since each sees only its own consultations.	Plan normalisation, structured diff, RxNorm equivalence	9/10	2
Prescription comprehension verifier	Checks immediately after the visit whether the patient understood dose, timing and duration, and corrects misunderstandings. Misunderstanding of instructions is well documented and almost never measured.	Prescription parsing, question generation, scoring	7/10	2
Follow-up necessity predictor	Predicts whether a return visit is actually required, avoiding unnecessary trips without missing necessary ones. Valuable where patients travel several hours to a clinic.	Gradient boosting over outcome table, calibration curve	7/10	2
Literacy-adapted multilingual instructions	Rewrites the treatment plan into the patient's language at an actionable reading level, with readability measured before and after. Prescriptions get translated but not simplified.	IndicTrans2, readability scoring, text-to-speech	6/10	2

9. Recommended build plan

Foundation repairs, required regardless of direction

REPAIR	WHY IT CANNOT BE SKIPPED	WEEKS
Conversation memory	The model currently sees one message at a time. This is a defect, not a missing feature, and it invalidates every claim about conversational ability.	1
Own tabular models with SHAP	Train gradient-boosted models on Pima, Cleveland and ILPD, then compute per-prediction SHAP values. Replaces hardcoded weights that will not survive examination.	2
API key moved server-side	The key is currently readable by any visitor to the site.	< 1
Adversarial red-team suite	A battery of prompts attempting to extract a diagnosis, drug name or dose, reporting refusal rate and emergency recall. Inexpensive and taken seriously by examiners.	1

Minimum set that tells the whole story

Five features, approximately eleven weeks, one coherent narrative spanning all three windows.

WINDOW	FEATURE	WEEKS	WHAT IT DEMONSTRATES
Before	Camera vitals and come-prepared engine	5	Signal processing plus retrieval, with an immediately obvious practical benefit
Between	Passive phenotyping and side-effect early warning	4	Time-series modelling on data already held, plus a real biomedical knowledge base
After	Causal effectiveness with doctor dashboard	3	Causal inference and the data flywheel argument, which is the part examiners remember

Start with the feature that needs no new data

Passive digital phenotyping runs entirely on message timestamps already being written to Firestore. Two weeks, no new integration, and the most research-flavoured item on the list. That makes it the cheapest possible proof that the direction works.

How the stages chain into a system

STEP	PRODUCES	WHICH ENABLES
Pre-visit intake	A structured baseline: complaint, onset, current medicines, camera vitals	Everything downstream, because a day-zero reference point now exists
Doctor brief and visit	The treatment plan in machine-readable form	Side-effect watch, comprehension check, plan comparison
Between-visit monitoring	A symptom and behaviour time series against a personal baseline	Deterioration alerts, trigger inference, recovery trajectory
Outcome capture	Whether treatment worked, measured per patient and per treatment	Causal effectiveness, follow-up prediction, doctor feedback loop
The loop closes	Outcomes retrain the models that generated the advice	A system that improves with use, which no incumbent can replicate

10. Datasets and evaluation metrics

Datasets

DATASET	USED FOR	ACCESS
Pima, Cleveland, ILPD	Tabular models the team trains and owns	Open, UCI repository
SIDER	Drug side-effect profiles for the early-warning feature	Open
RxNorm and open DrugBank subset	Drug normalisation and interaction graph	Free tier or registration
WHO, ICMR and NICE guidelines	Retrieval corpus, the grounding backbone	Public documents
Coswara (IISc)	Indian respiratory audio, if voice biomarkers are pursued	Open
OpenAQ or CPCB feeds	Air quality history for trigger inference	Open API
MedQuAD	Question-answer pairs for evaluation	Open
MIMIC-IV	Realistic patient context, optional	Credentialed; begin the process early

Metrics

The objective is not to outperform Ada Health. It is to be the only project that measured itself at all.

METRIC	PUBLISHED BASELINE	TARGET
Top-3 triage accuracy	Ada 76%, K Health 74%	Report honestly; exceeding it is not required
Emergency recall	Ada 18 of 20	Report per vignette; this is the safety-critical figure
Citation precision	Published models: 3.6–29.7%	Above 90% by construction, via retrieved-span linking
Unsupported claim rate	Not reported by any consumer application	Measure it; being first to report is itself the contribution

METRIC	PUBLISHED BASELINE	TARGET
Must-ask question coverage	Not reported	Track before and after the active questioning loop
Adversarial refusal rate	Not reported	Above 95% on the red-team battery

11. Positioning and sources

The positioning statement

"The industry has optimised the seven minutes a patient spends with a doctor. We instrument the hundred and fifty days around it. We collect a structured baseline before the visit, including vitals measured through the phone camera. We watch for side effects and deterioration against that patient's own baseline rather than a population range. We reconcile prescriptions across every doctor they see, which an EMR vendor structurally cannot do because it only sees its own network. And we are the only ones who ask afterwards whether the treatment actually worked. Every statement the system makes is audited against retrieved clinical guidelines, and we report our own unsupported-claim rate — a number no consumer health product publishes. It is not a competing EMR. It is the patient-side intelligence layer that no EMR can build."

Terminology to avoid

Do not describe the project as a symptom checker or a chatbot. Both terms map it onto a commoditised category in the listener's mind before the sentence finishes. Lead with simulation, reconciliation and verification instead.

Sources for the literature review

WORK	RELEVANCE
CPGBench — clinical practice guideline detection and adherence in multi-turn conversations	The core justification. Establishes the detection-versus-citation gap across eight leading models and 32,155 recommendations.
MedRAGChecker — claim-level verification for biomedical retrieval-augmented generation	Blueprint for the claim verifier: decompose an answer into claims, verify each against retrieved evidence, aggregate to answer level.
MedDialogRubrics — benchmark for multi-turn medical consultations	Blueprint for the evaluation harness: a simulated patient agent anchored to atomic facts, with must-ask rubric scoring.
ClinHallu — stage-wise hallucination diagnosis in medical multimodal reasoning	Argues for decomposing the pipeline into visual recognition, knowledge recall and reasoning integration so failures can be localised rather than merely counted.

WORK	RELEVANCE
MediEval — patient-contextual and knowledge-grounded reasoning benchmark	Safety framing: documents hallucinated support and truth inversion, and shows targeted fine-tuning can eliminate them.
AI symptom checker accuracy review, 2026	Source of the comparative accuracy and emergency-recognition figures for Ada, K Health, Babylon, Buoy, Isabel and WebMD.
India EHR market report, 2026–2032	Source of HealthPlix, Practo and Eka Care scale and positioning data.

Generated from a full audit of the MediAI repository. All code references correspond to file paths and line numbers in the repository at the time of review.